

Hao Chen

CONTACT INFORMATION

West Lafayette, IN, USA
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EDUCATION

Purdue University, West Lafayette, IN
Ph.D. in Chemical Engineering 2022 - Present
• Research advisor: Dr. [Can Li](#)
• Research expertise: Machine Learning and Mathematical Optimization

University of Cambridge, Cambridge, UK
M.Phil. in Chemical Engineering and Biotechnology 2020 - 2021

University of Nottingham, Nottingham, UK
B.Eng. in Chemical Engineering 2016 - 2020

APPOINTMENTS

Amazon, Bellevue, WA 2025
Applied Scientist Intern, Last Mile Planning Supervisor: Dr. Jin Ye

PUBLICATIONS

- [1] GE Constante-Flores, **H Chen**, C Li. Enforcing Hard Linear Constraints in Deep Learning Models with Decision Rules. *The Thirty-ninth Annual Conference on Neural Information Processing Systems* (2025).
- [2] A Ramanujam, A Elyoumi, H Chen, SM Kompalli, AS Ahluwalia, S Pal, DJ Papageorgiou, C Li. SafeOR-Gym: A Benchmark Suite for Safe Reinforcement Learning Algorithms on Practical Operations Research Problems. *arXiv*. (2025).
- [3] H Kim, H Chen, C Li, JM Lee. TalkToAgent: A Human-centric Explanation of Reinforcement Learning Agents with Large Language Models. *arXiv*. (2025).
- [4] **H Chen**, GE Constante-Flores, KSI Mantri, SM Kompalli, AS Ahluwalia, C Li. OptiChat: Bridging Optimization Models and Practitioners with Large Language Models. *INFORMS Journal on Data Science*. (2025).
- [5] A Khan, R Nahar, H Chen, GE Constante-Flores, C Li. FaultExplainer: Leveraging Large Language Models for Interpretable Fault Detection and Diagnosis. *Computers & Chemical Engineering*, 109152. (2024).
- [6] YC Lu, A Varghese, R Nahar, H Chen, K Shao, X Bao, C Li. scChat: A Large Language Model-Powered Co-Pilot for Contextualized Single-Cell RNA Sequencing Analysis. *bioRxiv*. (2024).
- [7] **H Chen**, GE Constante-Flores, C Li. Physics-informed neural networks with hard linear equality constraints. *Computers & Chemical Engineering* 189, 108764. (2024).
- [8] **H Chen**, GE Constante-Flores, C Li. Diagnosing Infeasible Optimization Problems Using Large Language Models. *INFOR: Information Systems and Operational Research*, 1–15. (2024).
- [9] Y Yan, WI Shin, H Chen, et al. A recent trend: application of graphene in catalysis. *Carbon Lett.* 31, 177–199 (2021).

CONFERENCE PRESENTATIONS

Chen, H. (2025). Deep Learning Models with Hard Linear Inequality Constraints. Paper presented at the 2025 *AIChE Annual Meeting*, Boston, MA (CAST Director's Student Presentation Award Finalist)

Chen, H. (2024). Self-supervised Learning for Constrained Optimization with Hard Linear Constraints. Paper presented at the 2024 *INFORMS Annual Meeting*, Seattle, WA

Chen, H. (2024). GPU Accelerated Approximation Algorithm for Multi-Parametric Linear Programming. Paper presented at the 2024 AIChE Annual Meeting, San Diego, CA

Chen, H. (2024). Physics-Informed Neural Networks with Hard Linear Equality Constraints. Paper presented at the 2024 AIChE Annual Meeting, San Diego, CA

Chen, H. (2024). Diagnosing Infeasible Optimization Problems Using Large Language Models. Paper presented at the 2024 AIChE Annual Meeting, San Diego, CA

HONORS AND AWARDS	Purdue Graduate Student Government (PGSG) Travel Grant	2025
	Ryland Travel Grant, School of Chemical Engineering, Purdue University	2025
	Purdue Graduate Student Government (PGSG) Travel Grant	2024
	Chao Travel Grant, School of Chemical Engineering, Purdue University	2024
	Nottingham Engineering Excellence Scheme (Top 1.0%), University of Nottingham	2019
	British Petroleum Prize, University of Nottingham	2019
	Provost's Scholarship (Top 1.5%), University of Nottingham	2018
	Dean's Scholarship (Top 10%), University of Nottingham	2017
RESEARCH MENTORING	Priya Kumari (M.S. from Purdue)	
	Ashley Jojan Varghese (M.S. from Purdue)	
	Zachary Rasmussen (Undergraduate from Utah)	
	Rahul Golder (Undergraduate from IIT)	
	Soumick Sarker (Undergraduate from IIT)	
	Shraman Pal (Undergraduate from IIT)	
PROFESSIONAL ACTIVITIES	Reviewer for <i>Digital Chemical Engineering</i>	2025
	Reviewer for <i>Applied Energy</i>	2025
	Reviewer for <i>Computer & Chemical Engineering</i>	2025
	Reviewer for <i>Journal of Aviation Technology and Engineering</i>	2025
	Reviewer for <i>ACS In Focus</i>	2024
	Reviewer for <i>Reviews in Chemical Engineering</i>	2024
TEACHING EXPERIENCE	Statistical Modeling and Quality Enhancement (CHE-32000), TA	Fall 2024
	Momentum Transfer (CHE-37700), TA	Fall 2023
SKILLS	• Programming: Python, Julia, MATLAB • Frameworks: PyTorch, PyG, SB3, Pyomo, OMLT, NumPy, Pandas, SciPy • Solvers: Gurobi, CPLEX, Xpress • Toolbox: Linux, vim, git	